Côme Hosxe

come.hosxe@gmail.com linkedin.com/in/comehosxe comoxx.github.io

Available immediately for full-time positions

Education

Master Mathématiques, Vision, Apprentissage (MVA), ENS Paris-Saclay 2025 – 2026
- **Courses:** Deep Learning, Generative Models, Graphical Models, Multimodal Explainable AI, Convex and Large-Scale Optimization, DL for Medical Imaging.

CentraleSupélec, Engineering degree (MSc), Université Paris-Saclay 2022 – 2026
- **Major:** Mathematics and Data Science. **GPA:** 4.33/4.33 (final year)
- **Courses:** Machine Learning, Reinforcement Learning, Computer Vision, Statistics, Stochastic Calculus, Harmonic Analysis.

Preparatory Class (MPSI, MP*), Lycée Louis-le-Grand, Paris 2020 – 2022

Professional Experience

Machine Learning Research Intern, Callyope, Paris May 2026 – Oct. 2026
- Research question: do reasoning audio language models (ALMs) ground their reasoning in the voice?
- Designed a benchmark that grades each acoustic claim of a model's reasoning against clinician ratings: 5 datasets (English, Spanish), 25 perceptual dimensions, 10 open-weight reasoning ALMs evaluated.
- Post-trained Qwen2.5-Omni (7B) with SFT on chain-of-thought traces, then GRPO with a reward I designed on claim correctness. Claim factuality went from 0.54 to 0.60.
- Multi-GPU training and evaluation on Jean Zay and Leonardo. First-author paper (ICASSP 2027 workshop).

Machine Learning Research Intern, WickerLab, Univ. of Auckland, NZ Mar. 2025 – Jun. 2025
- Built a live bird detection pipeline, from a forest microphone to a museum art installation (2026).
- Fine-tuned BirdNET on New Zealand species with a 2,000-clip dataset from 3 sources, filtered by SNR and a teacher model. Unlike stock BirdNET, it does not predict non-native species on field recordings.
- Rewrote the recorder software (Raspberry Pi, 4G) for live WebSocket streaming and built a React/Django platform for researchers.

Data Scientist Intern, Datamaran, Valencia, Spain Jul. 2024 – Dec. 2024
- RAG pipeline that suggests ESG impacts, risks and opportunities from corporate reports: embedding retrieval (incl. Matryoshka), UMAP/HDBSCAN theme clustering, LLM generation per cluster.
- Analysed cluster stability, sped up the pipeline, built semantic deduplication and a Streamlit app for experts.

Teaching Assistant, CentraleSupélec Sept. 2023 – Nov. 2023
- Taught lectures and tutorials to 30 first-year students: Bash, Git, OOP, SQL, networks and computer security.

Publications

- **First author**, *SLP-Bench: Evaluating and Improving the Voice Claims of Reasoning Audio Language Models on Clinical Speech*. Submitted to an ICASSP 2027 workshop.
- Co-author (L2S), *Delivery Race Game: a Stimulative Approach to Engage Youths in Robotics and Control*. IFAC WACE 2024.
- Co-author (L2S), *DroPong: Revisiting the Pong Game with Aerial and Ground Drones*. European Control Conference 2024.

Projects and Activities

AI for Text-Based Deduction, CentraleSupélec Oct. 2025 – Mar. 2026
- AI agent for the word-guessing game Pédantix (RL, language models, clustering), with Illuin Technology.

President of Kampuchea Souriya, Volunteer Association Dec. 2022 – Dec. 2023
- Led a student solidarity association working in Cambodia, raised over $30,000 and built its website.

Skills

ML / LLM: PyTorch, Hugging Face Transformers, LoRA, vLLM, GRPO, Scikit-learn, OpenCV.

Engineering: Python, SQL, Bash, Git, SLURM, HPC.

Languages: French (native), English (C1), Spanish (B1), German (B1).

Hobbies: Freediving, hiking and outdoor activities, sudoku.